

Math 344, F18
Quiz 7

Name: Answer

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Let $A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 6 & 4 & 3 \\ -2 & -6 & 0 & -1 \end{bmatrix}$.

(a) Transform A into the reduced echelon form R .

$$A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 2 & 6 & 4 & 3 \\ -2 & -6 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 0 & \frac{1}{2} \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

$\underbrace{\hspace{1.5cm}}_{\text{pivot variables}} \quad \underbrace{\hspace{1.5cm}}_{\text{free variables}}$

(b) Find the dimension and a basis for the column space of A , $C(A)$.

$$\dim C(A) = 2$$

$$\text{Basis: } \left\{ \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} \right\}$$

(c) Find the dimension and a basis for the row space of A , $C(A^T)$.

$$\dim C(A^T) = 2$$

$$\text{Basis: } \left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix} \right\}$$

(d) Find the dimension and a basis for the nullspace of $\mathbf{A}$, $N(\mathbf{A})$.

$$\dim(N(\mathbf{A})) = 4 - 2 = 2$$

$$R\mathbf{x} = 0 \Rightarrow \begin{cases} x_1 + 3x_2 + \frac{1}{2}x_4 = 0 \\ x_3 + \frac{1}{2}x_4 = 0 \end{cases}$$

$$x_2 = 1, x_4 = 0 \Rightarrow S_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \text{Basis: } \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \\ \frac{1}{2} \end{bmatrix} \right\}$$

$$x_2 = 0, x_4 = 1 \Rightarrow S_2 = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \\ \frac{1}{2} \end{bmatrix}$$

(e) What is the dimension of the left nullspace of $\mathbf{A}$, $N(\mathbf{A}^T)$?

$$\dim(N(\mathbf{A}^T)) = 3 - 2 = 1$$